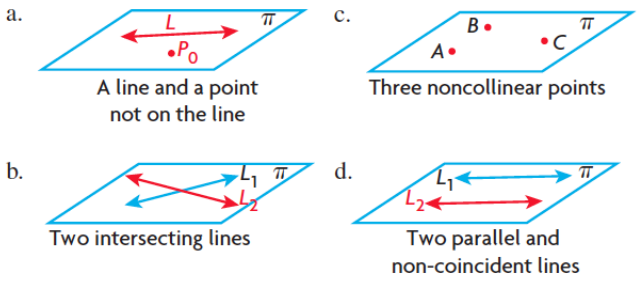
L3 – Equations of Planes Unit 6
MCV4U
Jensen

To determine the equation of a plane you need to know two non-collinear vectors parallel to the plane and a point on the plane. We can determine this information if any of the following information is given:

https://www.geogebra.org/3d/hpnm54gs



The equation of a plane can be written in the following forms:

i) Vector
ii) Parametric
iii) Scalar (next lesson)

Vector Equation of a Plane	Parametric Equation of a Plane
$\vec{r} = \overrightarrow{r_0} + t\vec{a} + s\vec{b}$ OR $[x, y, z] = [x_0, y_0, z_0] + t[a_1, a_2, a_3] + s[b_1, b_2, b_3]$ $\vec{r} = [x, y, z]$ is a position vector for any point on the plane. $\overrightarrow{r_0} = [x_0, y_0, z_0]$ is a position vector for a known point on the plane. $\vec{a} = [a_1, a_2, a_3]$ and $\vec{b} = [b_1, b_2, b_3]$ are non-parallel direction vectors parallel to the plane. t and s are scalars	$\begin{cases} x = x_0 + ta_1 + sb_1 \\ y = y_0 + ta_2 + sb_2 \\ z = z_0 + ta_2 + sb_2 \end{cases}$

Example 1: Consider the plane with direction vectors $\vec{a} = [8, -5, 4]$ and $\vec{b} = [1, -3, -2]$ through $P_0(3,7,0)$.

https://www.geogebra.org/3d/s44t62d7

a) Write the vector and parametric equations of the plane.

Vector Equation:

$$[x, y, z] = [3,7,0] + t[8, -5, 4] + s[1, -3, -2]$$

Parametric Equations:

$$\pi: \begin{cases} x = 3 + 8t + s \\ y = 7 - 5t - 3s \\ z = 4t - 2s \end{cases}$$

b) Determine if the point $Q(-10,8,-6)$ is on the plane.

If the point is on the plane, then there exists a single set of t and s values that satisfy the equations.

① $-10 = 3 + 8t + s$

② $8 = 7 - 5t - 3s$

③ $-6 = 4t - 2s$

Solve for s and t using any two equations:

$3 \times$ ① $\quad -30 = 9 + 24t + 3s$

② $\quad 8 = 7 - 5t - 3s \quad +$

$-22 = 16 + 19t$

$t = -2$

Sub $t = -2$ into ① or ② to solve for s

$-10 = 3 + 8(-2) + s$

$s = 3$

Now Check to see if $t = -2$ and $s = 3$ also satisfy ③:

LS

$= -6$

RS

$= 4(-2) - 2(3)$

$= -14$

$LS \neq RS$

Therefore point $Q(-10,8,-6)$ is NOT on the plane.

c) Find the coordinates of two other points on the plane.

Choose arbitrary values for parameters t and s:

Let $t = 1$ and $s = -1$

$[x, y, z] = [3,7,0] + 1[8, -5,4] - 1[1, -3, -2]$

$[x, y, z] = [10,5,6]$

Let $t = 2$ and $s = 1$

$[x, y, z] = [3,7,0] + 2[8, -5,4] + 1[1, -3, -2]$

$[x, y, z] = [20, -6,6]$

Two points on the plane are $(10,5,6)$ and $(20, -6,6)$.

d) Find the x-intercept of the plane.

Set $y = z = 0$ and solve for s and t:

① $x = 3 + 8t + s$

② $0 = 7 - 5t - 3s$

③ $0 = 4t - 2s$

$2 \times$② $\quad 0 = 14 - 10t - 6s$

$3 \times$③ $\quad 0 = 12t - 6s \quad -$

$0 = 14 - 22t$

$t = \frac{7}{11}$

Sub $t = \frac{7}{11}$ in to ② or ③ and solve for s:

$0 = 4\left(\frac{7}{11}\right) - 2s$

$2s = \frac{28}{11}$

$s = \frac{14}{11}$

Now sub $t = \frac{7}{11}$ and $s = \frac{14}{11}$ in to ① and solve for x:

$x = 3 + 8\left(\frac{7}{11}\right) + \frac{14}{11}$

$x = \frac{33}{11} + \frac{56}{11} + \frac{14}{11}$

$x = \frac{103}{11}$

The x-intercept is $\left(\frac{103}{11}, 0,0\right)$

Example 2: Find the vector and parametric equations of each plane.

a) The plane with x-intercept = 2, y-intercept = 4, and z-intercept = 5.

Determine 2 direction vectors:

$\vec{a} = \overrightarrow{XY}$

$\vec{a} = [0,4,0] - [2,0,0]$

$\vec{a} = [-2,4,0]$

$\vec{b} = \overrightarrow{XZ}$

$\vec{b} = [0,0,5] - [2,0,0]$

$\vec{a} = [-2,0,5]$

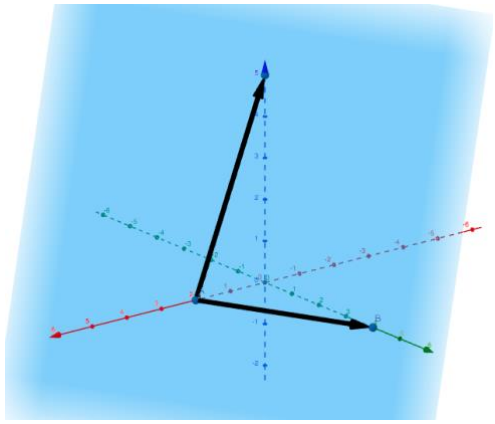
Vector Equation:

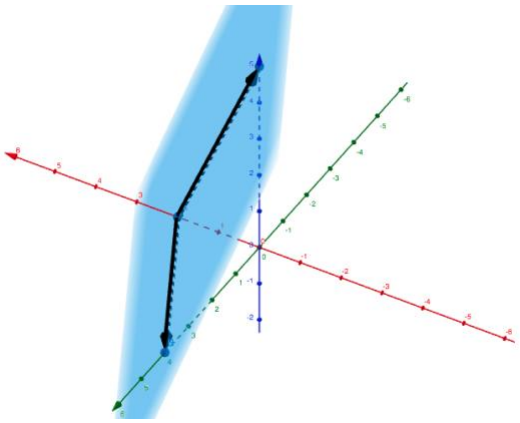
$[x, y, z] = [2,0,0] + t[-2,4,0] + s[-2,0,5]$

Parametric Equations:

$$\pi: \begin{cases} x = 2 - 2t - 2s \\ y = 4t \\ z = 5s \end{cases}$$

3D visualization: https://www.geogebra.org/3d/czgh5pk5





b) The plane containing the line $[x, y, z] = [0,3,-5] + t[6,-2,-1]$ and parallel to the line $[x, y, z] = [1,7,-4] + s[1,-3,3]$.

Point $(0,3,-5)$ is on the plane so a position vector is $[0,3,-5]$

Two direction vectors are $[6,-2,-1]$ and $[1,-3,3]$. Note that these are not scalar multiple multiples of each other and therefore are not parallel.

Vector Equation:

$$[x, y, z] = [0,3,-5] + t[6,-2,-1] + s[1,-3,3]$$

Parametric Equations:

$$\pi: \begin{cases} x = 6t + s \\ y = 3 - 2t - 3s \\ z = -5 - t + 3s \end{cases}$$

Geogebra: https://www.geogebra.org/3d/xskun4uf